WEEK 7: STATISTICS II

1. INTRODUCTION

Last week, we organized raw data into simple frequency tables.

But when data is **continuous or very large**, we group it into **class intervals**.

We also represent such data using:

- Bar charts
- Histograms

Why Group Data?

Ungrouped data (example):
150, 151, 152, 153, 154, 155...
❌ Too many values
❌ Hard to understand

Grouped data:

- 150–154 → 15 people
- 155–159 → 25 people
- 160–164 → 35 people

✔ Clear pattern
✔ Easy to interpret

2. CONTENT

A. FREQUENCY DISTRIBUTION (GROUPED DATA)

1. Class Interval

A class interval is a **range used to group data**.

Examples:

- 10–19, 20–29, 30–39
- 0–4, 5–9, 10–14

Visual

```
10–19   [====]
20–29   [====]
30–39   [====]
40–49   [====]
```

2. Class Boundaries

To remove gaps between intervals:

Rule:

- Subtract 0.5 from lower limit
- Add 0.5 to upper limit

Examples:

- 10–19 → 9.5–19.5
- 20–29 → 19.5–29.5

Visual Explanation

Stated: 10–19 20–29 30–39
Boundaries: 9.5–19.5–29.5–39.5

No gaps → Continuous data

3. Class Width

Class Width = Upper Boundary − Lower Boundary

Example:
19.5 − 9.5 = 10

4. Class Midpoint

Midpoint = (Lower Limit + Upper Limit) ÷ 2

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Example:
10–19 → (10 + 19) ÷ 2 = 14.5

Visual

10 -------- 19
↓
14.5

COMPREHENSIVE EXAMPLE

Raw data (30 scores):

5, 8, 6, 7, 7, 10, 9, 4, 5, 6,
7, 8, 9, 7, 6, 5, 8, 7, 7, 9,
4, 6, 8, 7, 6, 7, 8, 9, 5, 7

Step 1: Range

Minimum = 4
Maximum = 10

Range = 10 − 4 = 6

Step 2: Class Intervals

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4–5
6–7
8–9
10–11

Step 3: Frequency Table

Class	Tally	Frequency
4–5		
6–7		
8–9		
10–11		
Total		30

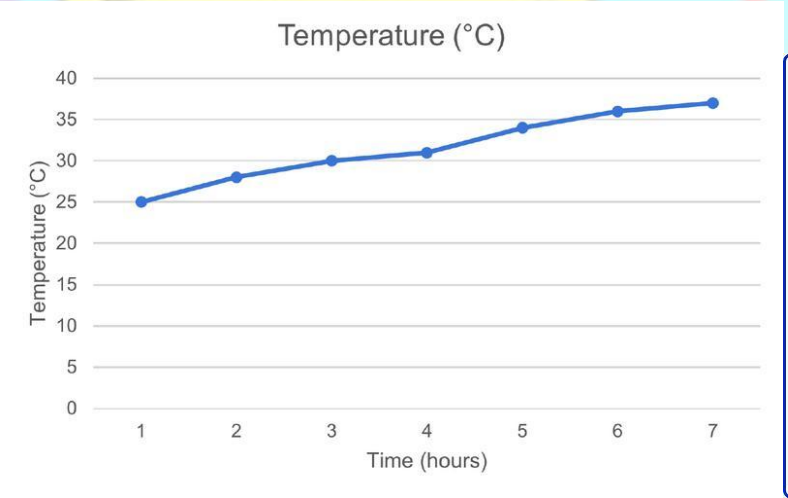
With Boundaries & Midpoints

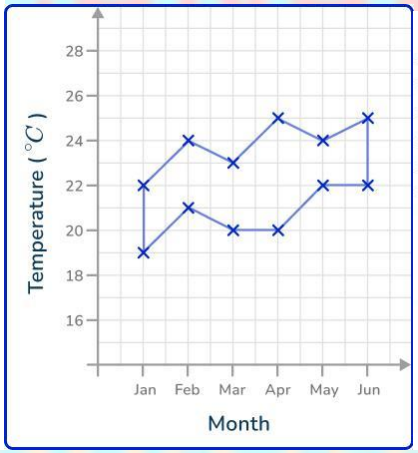
Class	f	Boundaries	Midpoint
4–5	4	3.5–5.5	4.5
6–7	14	5.5–7.5	6.5
8–9	9	7.5–9.5	8.5
10–11	1	9.5–11.5	10.5

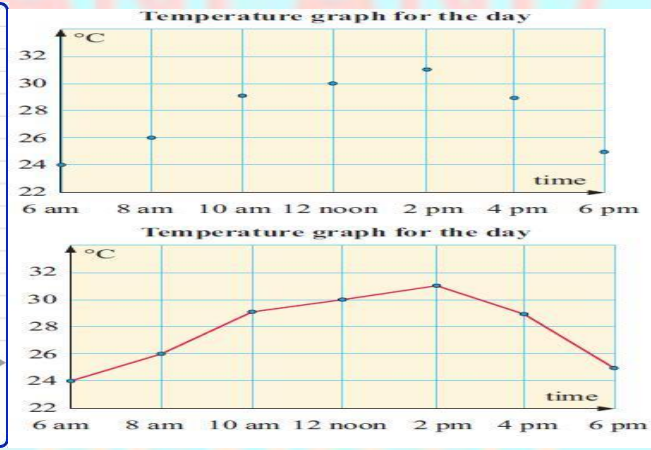
B. GRAPHS

1. LINE GRAPH

Used to show **trend over time**







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Features:

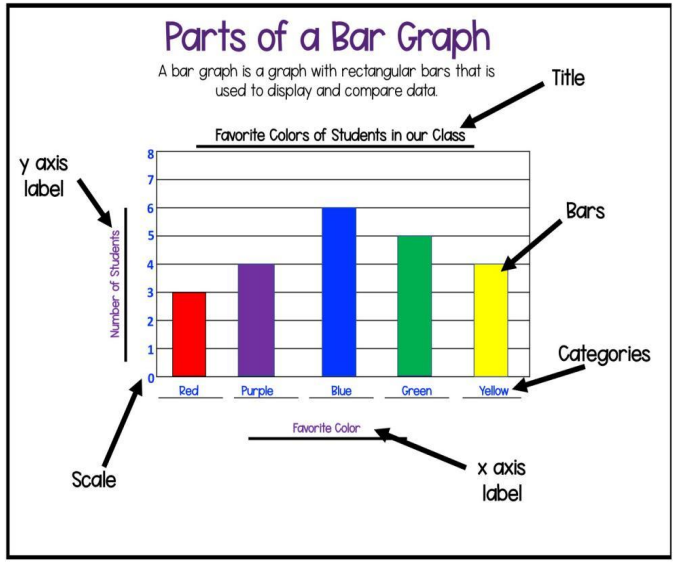
- Points connected with lines

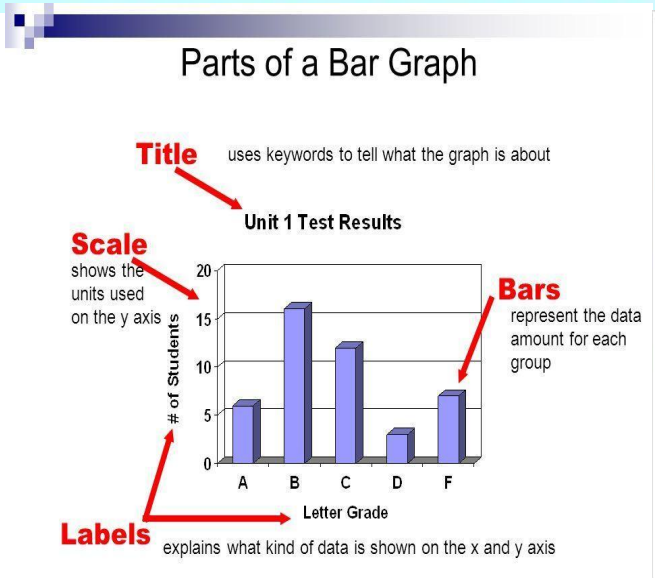
- Shows continuous change

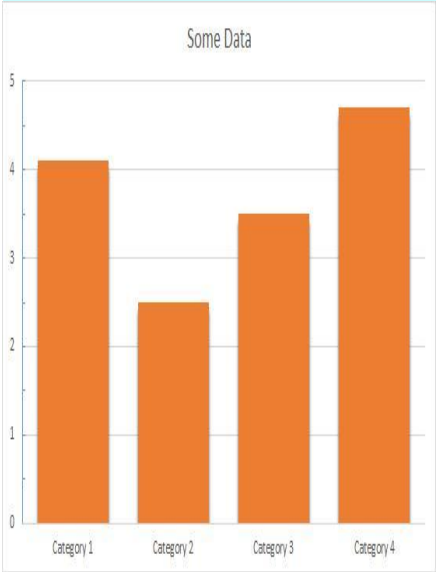
2. BAR CHART

Used for:

- Discrete data
- Categories







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Features:

- Has gaps between bars
- Equal width bars
- X-axis = categories
- Y-axis = frequency

3. HISTOGRAM

Used for:

- Continuous data

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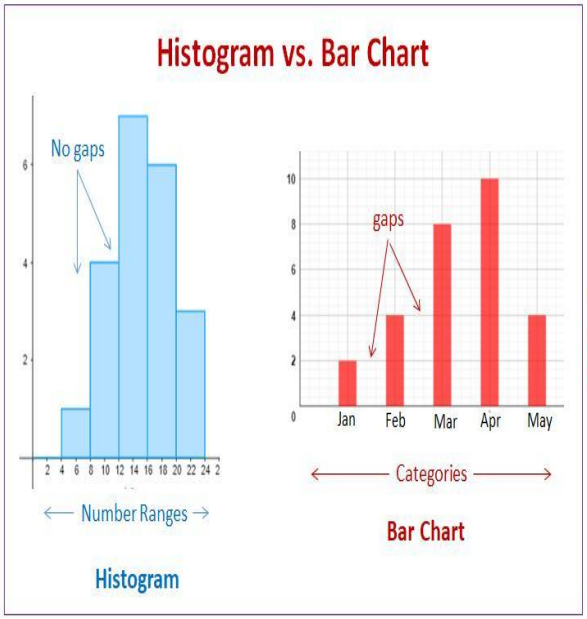

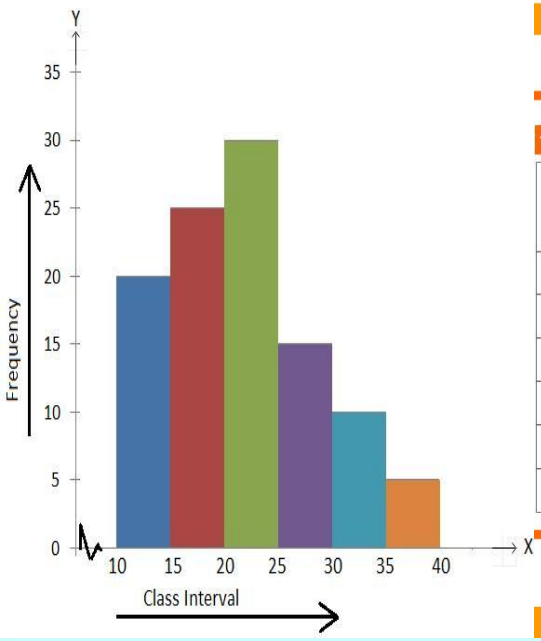

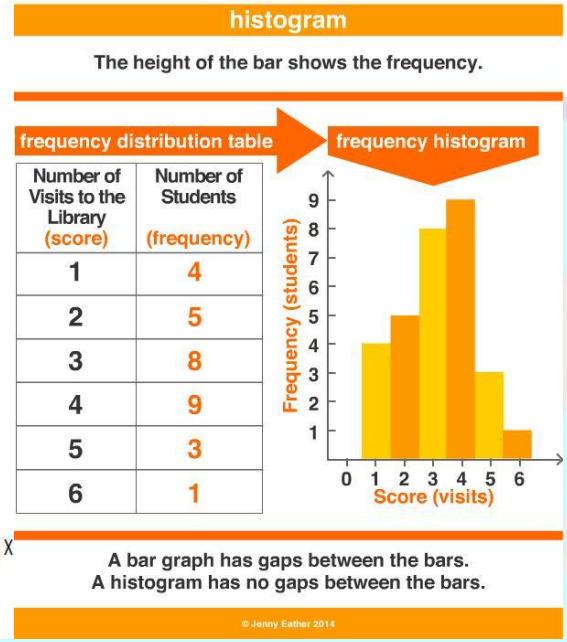

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Features:

- Bars touch (no gaps)
- X-axis = class boundaries
- Y-axis = frequency

Bar Chart vs Histogram

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Feature	Bar Chart	Histogram
Data Type	Discrete	Continuous
Gaps	YES	NO
X-axis	Categories	Boundaries

3. WORKED EXAMPLE (HISTOGRAM)

Example 9.1

Heights of 50 students:

Height (cm)	Frequency
150–154	6
155–159	12
160–164	18
165–169	10
170–174	4

Step 1: Class Boundaries

Interval	Boundaries
150–154	149.5–154.5

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Interval Boundaries

155–159 154.5–159.5
160–164 159.5–164.5
165–169 164.5–169.5
170–174 169.5–174.5

Step 2: Axes

- X-axis → boundaries
- Y-axis → frequency

Step 3: Draw Histogram

- Draw bars touching each other
- Heights = frequencies

Real Histogram

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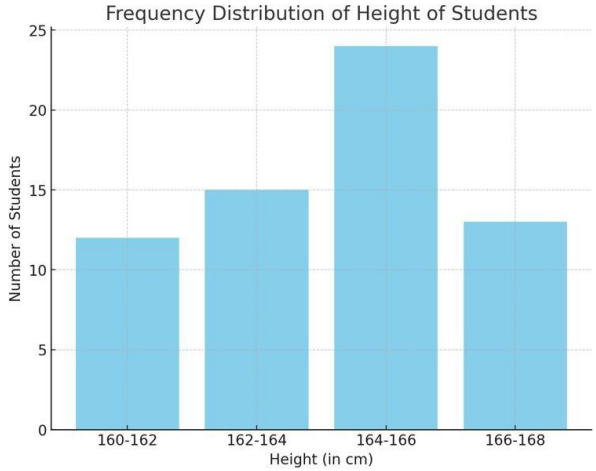

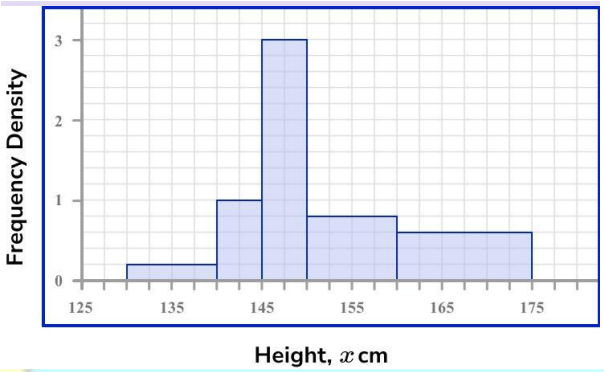

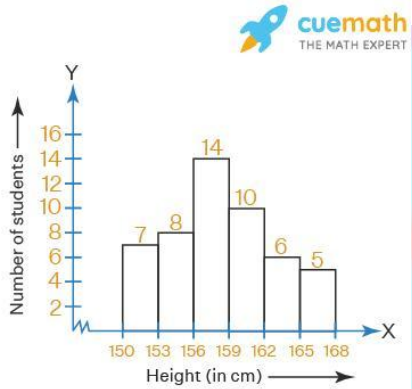

Interpretation

- Highest class = 160–164 (modal class)
- Fewest = 170–174
- Shape ≈ normal distribution

4. REVISION QUESTIONS

1. Why do we group data?
2. Difference between bar chart and histogram?
3. What data uses bar chart?
4. What data uses histogram?
5. What is on x-axis of histogram?

5. ASSESSMENT

Classwork

1. For 20–29:
   - midpoint
   - boundary
2. Meaning of gaps in bar chart
3. Use of line graph
4. Bar chart or histogram (children per family?)
5. Heights data → which graph?

Assignment

Given ages:

25, 30, 18, 45, 42, 28, 19, 20, 22, 33,
35, 40, 38, 29, 27, 22, 25, 30, 31, 26,
41, 37, 26, 28, 30, 33, 39, 21, 19, 44

Tasks:

a) Create grouped table using:

15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49

b) Draw histogram using boundaries

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